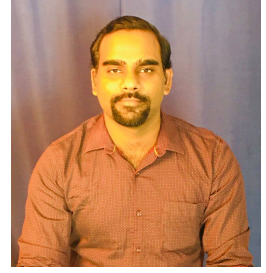


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List of Sponsored Projects

1. Physics of Neutron Stars: Constraints from Finite Nuclei to Gravitational Waves
PI: Dr Bharat Kumar (NIT, Rourkela)
Co-PI: Prof B K Agrawal (SINP, Kolkata)
Sponsoring agency: SERB
Tenure of the Assignment: 2022-2025
Total Amount: 21,12,828 Rs.

Research Publications

II. Preprinted Papers

1. Structure effects on fission yields,
Bharat Kumar, M. T. Senthil Kannan, M. Balasubramaniam, B. K. Agrawal, S. K. Patra,
[<https://arxiv.org/abs/1701.00731>]
2. Observable Signatures of Quarkyonic Phase in Neutron Stars,
Probit J Kalita, Tuhin Malik, Tianqi Zhao, Bharat Kumar, James M. Lattimer,
Prepare for PRL[<https://arxiv.org/abs/2510.23405>]

Chapters in book

1. Recent parametrization in relativistic mean-field formalism,
K C Naik, R. N. Panda, **Bharat Kumar** and S. K. Patra,
Nuclear structure physics, CRC press (2020) ISBN:9780367256104

Papers Published in International Refereed Journals

1. Universal Relations and Correlation Analysis of Proto-Neutron Star Properties in Energy-Momentum Squared Gravity,
Sayantan Ghosh,
[arXiv:2602.02069 \(2026\)](#)
2. Trace anomaly and interior curvature of neutron stars in energy-momentum squared gravity,
Ratikanta Swain, Sayantan Ghosh, **Bharat Kumar**,
[arXiv:2606.20203 \(2026\)](#)
3. Constraining the f-mode oscillations frequency in neutron stars through universal relations in the realm of energy-momentum squared gravity,
Sayantan Ghosh,
Gen. Relativ. Gravit. 57 (2025) 70 [<https://arxiv.org/abs/2412.20815>]
Impact Factor:1.5
4. Spacetime Curvature as a Probe of Exotic Core Phases in Neutron Stars within Modified Gravity,
Sayantan Ghosh, **Bharat Kumar**, Subhash Mahapatra,
Phys. Rev. D 113 (2026) 024070 [<https://arxiv.org/pdf/2508.08866>]

Impact Factor:5.30

5. Astrophysical constraints on neutron star f-modes with a nonparametric equation of state representation,
Sailesh Ranjan Mohanty, Utkarsh Mali, H.C. Das, **Bharat Kumar**, Philippe Landry,
[Phys. Rev. D 112 \(2025\) 123043](#) [<https://arxiv.org/abs/2410.16689>]
Impact Factor:5.30
6. Characterizing the nuclear models informed by PREX and CREX: a view from Bayesian inference,
Tianqi Zhao, Zidu Lin, **Bharat Kumar**, Andrew W. Steiner, Madappa Prakash,
[Phys. Rev. Research 7 \(2025\) 043335](#) [<https://arxiv.org/abs/2406.05267>]
Impact Factor:4.2
7. The role of adiabatic sound speeds in neutron star radial oscillations and stability,
Sayantan Ghosh, Tianqi Zhao, **Bharat Kumar**, Sailesh Ranjan Mohanty,
[JCAP 09 \(2025\) 025](#) [<https://doi.org/10.1088/1475-7516/2025/09/025>]
Impact Factor:7.28
8. Decoding dark matter admixed neutron stars: From static structure to rotational deformation,
Pinku Routaray, Abirbhav Chakrawarty, N. K. Patra, **Bharat Kumar**,
[Phys. of the Dark Universe 50 \(2025\) 102093](#) [<https://arxiv.org/abs/2409.02131>]
Impact Factor:6.4
9. Effects of asymmetric dark matter on a magnetized neutron star: A two-fluid approach,
Pinku Routaray, Vishal Parmar, H. C. Das, **Bharat Kumar**, G. F. Burgio, and H.-J. Schulze,
[Phys. Rev. D 111 \(2025\) 103045109](#) [<https://arxiv.org/abs/2412.21097>]
Impact Factor:5.30
10. Correlation between the curvature and some properties of the neutron star,
S K Biswal, H C Das, Ankit Kumar, **Bharat Kumar**, Rashmita Jena, Padmalaya Dash and S K Patra,
[IJMP E 34 \(2025\) 2550009](#). [<https://arxiv.org/abs/2012.13673>]
Impact Factor:1.174
11. Dark Matter Admixed Neutron Star in the light of HESS J1731-347 and PSR J0952-0607,
Pinku Routaray, H C Das, Jeet Amrit Pattnaik, and **Bharat Kumar**
[IJMP E 33 \(2024\) 2450052](#). [<https://arxiv.org/abs/2307.12748>]
Impact Factor:1.174
12. Exploring the impact of the σ -cut scheme on neutron stars oscillations and thermal relaxations,
Probit J Kalita, Pinku Routaray, Khokan Singha, **Bharat Kumar**, and B.K. Agrawal
[Phys. Rev. C 110 \(2024\) 045807](#) [<https://arxiv.org/abs/2308.09008>]
Impact Factor:3.4
13. Exploring the macroscopic properties of proto-neutron stars: Effects of entropy and lepton fraction,
Sayantan Ghosh, Shahebaj Shaikh, Probit J Kalita, Pinku Routaray, **Bharat Kumar**, and B.K. Agrawal
[Nucl. Phys. B 1008 \(2024\) 116697](#) [<https://arxiv.org/abs/2307.06892>]
Impact Factor:3.04
14. Unstable Anisotropic Neutron Stars: Probing the Limits of Gravitational Collapse,
S. R. Mohanty, Sayantan Ghosh, and **Bharat Kumar**,
[Phys. Rev. D 109 \(2024\) 123039](#) [<https://arxiv.org/abs/2304.02439>]
Impact Factor:5.30
15. Probing the impact of Delta-Baryons on Nuclear Matter and Non-Radial Oscillations in Neutron Stars,
Probit Jyoti Kalita, Pinku Routaray, Sayantan Ghosh, and **Bharat Kumar**, Bijay K. Agrawal
[JCAP 04 \(2024\) 065](#) [<https://arxiv.org/abs/2308.09008>] Impact Factor:7.28
16. The Impact of Anisotropy on Neutron Star Properties: Insights from I-f-C Universal Relations,

S. R. Mohanty, Sayantan Ghosh, Pinku Routaray, H. C. Das and **Bharat Kumar**,
JCAP 03 (2024) 054 [Invited Article] [<https://arxiv.org/abs/2305.15724>]
Impact Factor:7.28

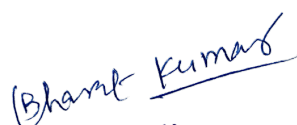
17. Investigating Dark Matter-Admixed Neutron Stars with NITR Equation of State in Light of PSR J0952-0607,
Pinku Routaray, Sailesh Ranjan Mohanty, H. C. Das, Sayantan Ghosh, P. J. Kalita, V. Parmar, **Bharat Kumar**,
JCAP 10 (2023) 073; [<https://doi.org/10.48550/arXiv.2211.12808>]
Impact Factor:7.28
18. Probing the impact of WIMP dark matter on universal relations, GW170817 posterior, and radial oscillations
Pinku Routaray, Abdul Quddus, Kabir Chakravarti, and **Bharat Kumar**,
MNRAS 525 (2023) 5492; [<https://arxiv.org/pdf/2202.04364.pdf>]
Impact Factor:5.23
19. Radial Oscillations of Dark Matter admixed Neutron Stars,
Pinku Routaray, H. C. Das, Souhardya Sen, **Bharat Kumar**, Grigoris Panotopoulos, and Tianqi Zhao,
Phys. Rev. D 107 (2023) 103039; [<https://doi.org/10.48550/arXiv.2211.12808>]
Impact Factor:5.30
20. Radial oscillations in neutron stars from unified equation of states,
Souhardya Sen, Shubham Kumar, Athul Kunjipurayil, Pinku Routaray, Sayantan Ghosh, Probit J. Kalita, Tianqi Zhao and **Bharat Kumar**,
Galaxies 11 (2023) 60; [Invited Article] [<https://arxiv.org/pdf/2205.02076.pdf>]
Impact Factor:3.3
21. Theoretical studies on structural properties and decay modes of $^{284-375}_{119}$ isotopes,
Asloob A. Rather, M. Ikram, Ishfaq A. Rather, A. A. Usmani, **Bharat Kumar**, K. P. Santhosh, S. K. Patra,
Eur. Phys. J. Plus 138 (2023) 467; [<https://arxiv.org/abs/1709.07311>]
Impact Factor:3.75
22. Impact of the equation of state on f - and p - mode oscillations of neutron stars,
Athul Kunjipurayil, Tianqi Zhao, **Bharat Kumar**, Bijay K. Agrawal, and Madappa Prakash,
Phys. Rev. D 106 (2022) 063005; [<https://arxiv.org/abs/2205.02081>]
Impact Factor:5.30
23. Dark Matter Effects on the Compact Star Properties,
H C Das, Ankit Kumar, **Bharat Kumar**, and S K Patra,
Galaxies 10 (2022) 14; [Invited Review Article]
Impact Factor:3.17
24. Universal relations between the quasinormal modes of neutron star and tidal deformability,
Hajime Sotani, and **Bharat Kumar**,
Phys. Rev. D 104 (2021) 123002; [<https://arxiv.org/pdf/2109.08145.pdf>]
Impact Factor:5.30
25. The BigApple force and it's implications to finite nuclei and astrophysical objects,
H C Das, Ankit Kumar, **Bharat Kumar**, S K Biswal, and S K Patra,
IJMP E 30 (2021) 2150088; [<https://arxiv.org/abs/2009.10690>]
Impact Factor:1.174
26. Impacts of dark matter on the curvature of the neutron star,
H C Das, Ankit Kumar, **Bharat Kumar**, S K Biswal, and S K Patra,
JCAP 01 (2021) 007; [<https://arxiv.org/abs/2007.05382>]
Impact Factor:5.21

27. Warm dense matter and cooling of supernovae remnants,
Ankit Kumar, H C Das, S K Biswal, **Bharat Kumar**, and S K Patra,
[Eur. Phys. J C 80 \(2020\) 775](https://doi.org/10.1140/epjc/s10052-020-8353-4); [https://doi.org/10.1140/epjc/s10052-020-8353-4]
Impact Factor: **4.39**
28. *GW170817 constraints on the properties of a neutron star in the presence of WIMP dark matter,
Abdul Quddus, Grigorios Panotopoulos, **Bharat Kumar**, Shakeb Ahmad, and S. K. Patra,
[J. Phys. G 47 \(2020\) 095202](https://doi.org/10.1088/1361-6471/ab9d36); [https://doi.org/10.1088/1361-6471/ab9d36]
Impact Factor: **2.41**
29. Effects of dark matter on the nuclear and neutron star matter,
H C Das, Ankit Kumar, **Bharat Kumar**, S K Biswal, Takashi Nakatsukasa, Ang Li, and S K Patra,
[MNRAS 495 \(2020\) 4893](https://doi.org/10.1093/mnras/staa1435); [https://doi.org/10.1093/mnras/staa1435]
Impact Factor: **5.54**
30. *Inferring Neutron Star Properties from GW170817 with Universal Relations
Bharat Kumar, and Philippe Landry,
[Phys. Rev. D 99 \(2019\) 123026](https://doi.org/10.1103/PhysRevD.99.123026); [https://doi.org/10.1103/PhysRevD.99.123026]
Impact Factor: **5.30**
31. ¹Constraints on the moment of inertia of PSR J0737-3039 from GW170817,
Philippe Landry, and **Bharat Kumar**,
[ApJL 868 \(2018\) L22](https://doi.org/10.3847/2041-8213/aaee76); [https://doi.org/10.3847/2041-8213/aaee76]
Impact Factor: **11.7**
32. *GW170817: Constraining the nuclear matter equation of state from the neutron star tidal deformability,
Tuhin Malik, N. Alam, M. Fortin, C. Providência, B. K. Agrawal, T. K. Jha, **Bharat Kumar**, and S. K. Patra,
[Phys. Rev. C 98 \(2018\) 034005](https://doi.org/10.1103/PhysRevC.98.035804); [https://doi.org/10.1103/PhysRevC.98.035804]
Impact Factor: **3.4**
33. Decay properties and reaction dynamics of zirconium isotopes in the relativistic mean-field model,
M. Panigrahi, R. N. Panda, **Bharat Kumar** and S. K. Patra,
[IJMP E 27 \(2018\) 1850012](https://doi.org/10.1142/S021830131850012X); [https://doi.org/10.1142/S021830131850012X]
Impact Factor: **1.174**
34. *New relativistic effective interaction for finite nuclei, infinite nuclear matter and neutron stars,
Bharat Kumar, S. K. Patra, and B. K. Agrawal,
[Phys. Rev. C 97 \(2018\) 045806](https://doi.org/10.1103/PhysRevC.97.045806); [https://doi.org/10.1103/PhysRevC.97.045806]
Impact Factor: **3.4**
35. A Study of Multi- Λ Hypernuclei within Spherical Relativistic Mean-field Approach,
Asloob A. Rather, M. Ikram, A. A. Usmani, **Bharat Kumar**, S. K. Patra,
[Braz. J. Phys. 47 \(2017\) 628](https://doi.org/10.1007/s13538-017-0525-9); [https://doi.org/10.1007/s13538-017-0525-9]
36. *New parameterization of the effective field theory motivated relativistic mean field model,
Bharat Kumar, S. K. Singh, B. K. Agrawal, S. K. Patra,
[Nucl. Phys. A 966 \(2017\) 197](https://doi.org/10.1016/j.nuclphysa.2017.07.001); [https://doi.org/10.1016/j.nuclphysa.2017.07.001]
Impact Factor: **1.70**
37. *Relative mass distributions of neutron-rich thermally fissile nuclei within statistical model,
Bharat Kumar, M. T. Senthil Kannan, M. Balasubramaniam, B. K. Agrawal, S. K. Patra,
[Phys. Rev. C 96 \(2017\) 034623](https://doi.org/10.1103/PhysRevC.96.034623); [https://doi.org/10.1103/PhysRevC.96.034623]
Impact Factor: **3.4**

¹Important papers which are related to the GW170817 and nuclear fission.

38. Relative fragmentation in ternary systems within the temperature-dependent relativistic mean-field approach,
M. T. Senthil Kannan, **Bharat Kumar**, M. Balasubramaniam, B. K. Agrawal, S. K. Patra,
[Phys. Rev. C 95 \(2017\) 064613](https://doi.org/10.1103/PhysRevC.95.064613); [https://doi.org/10.1103/PhysRevC.95.064613]
Impact Factor: **3.4**
39. *Tidal deformability of neutron and hyperon star with relativistic mean field equations of state,
Bharat Kumar, S. K. Biswal and S. K. Patra,
[Phys. Rev. C 95 \(2017\) 015801](https://doi.org/10.1103/PhysRevC.95.015801); [https://doi.org/10.1103/PhysRevC.95.015801]
Impact Factor: **3.4**
40. Structural and decay properties of $Z=132,138$ superheavy nuclei,
Asloob A. Rather, M. Ikram, A. A. Usmani, **Bharat Kumar**, S. K. Patra,
[Eur. Phys. J. A 52 \(2016\) 372](https://doi.org/10.1140/epja/i2016-16372-x); [https://doi.org/10.1140/epja/i2016-16372-x]
Impact Factor: **2.80**
41. Quest for magicity in hypernuclei,
M. Ikram, **Bharat Kumar**, S. K. Biswal and S. K. Patra,
[IJMP E 25 \(2016\) 1650103](https://doi.org/10.1142/S0218301316501032); [https://doi.org/10.1142/S0218301316501032]
Impact Factor: **1.174**
42. Effects of isovector scalar meson on neutron star both with and without hyperon,
S. K. Biswal, **Bharat Kumar** and S. K. Patra,
[IJMP E 25 \(2016\) 1650090](https://doi.org/10.1142/S0218301316500907); [https://doi.org/10.1142/S0218301316500907]
Impact Factor: **1.174**
43. Nuclear structure and decay properties of even-even nuclei in $Z = 70-80$ drip-line region,
S. Mahapatro, C. Lahiri, **Bharat Kumar**, R. N. Mishra and S. K. Patra,
[IJMP E 25 \(2016\) 1650062](https://doi.org/10.1142/S0218301316500622); [https://doi.org/10.1142/S0218301316500622]
Impact Factor: **1.174**
44. Modes of decay in neutron-rich nuclei,
Bharat Kumar, S. K. Biswal, S. K. Singh, Chirashree Lahiri, and S. K. Patra,
[IJMP E 25 \(2016\) 1650020](https://doi.org/10.1142/S0218301316500208); [https://doi.org/10.1142/S0218301316500208]
Impact Factor: **1.174**
45. Examining the stability of thermally fissile Th and U isotopes,
Bharat Kumar, S. K. Biswal, S. K. Singh and S. K. Patra,
[Phys. Rev. C 92 \(2015\) 054314](https://doi.org/10.1103/PhysRevC.92.054314); [https://doi.org/10.1103/PhysRevC.92.054314]
Impact Factor: **3.4**
46. Properties of superheavy nuclei: $Z = 124$,
M. S. Mehta, Harvinder Kaur, **Bharat Kumar** and S. K. Patra,
[Phys. Rev. C 92 \(2015\) 054305](https://doi.org/10.1103/PhysRevC.92.054305); [https://doi.org/10.1103/PhysRevC.92.054305]
Impact Factor: **3.4**
47. Shape co-existence and parity doublet in Zr isotopes,
Bharat Kumar, S. K. Singh and S. K. Patra,
[IJMP E 24\(2015\)1550017](https://doi.org/10.1142/S0218301315500172); [https://doi.org/10.1142/S0218301315500172]
Impact Factor: **1.174**

Sincerely,


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